From where heads look to what the stream carries

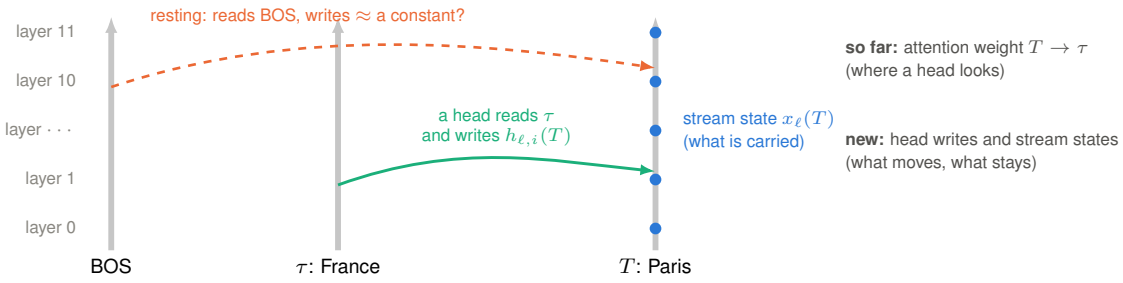
Notes on a topological analysis of the residual stream

Discussion notes for the next set of analyses. GPT-2 small. Builds on Exp 4 / 4b (a country at position τ; the final token T is its capital, a wrong capital or a random token) and the four attention-level topology analyses. Proposals, not results.

The shift in one paragraph. Everything so far measures *attention*: how much a head at T looks at the country at τ. Attention says where information could come from, not what is moved or kept. Each layer reads the residual stream, and every head and MLP adds an update to it:

$$x_{\ell+1}(T) \;=\; x_\ell(T) \;+\; \textstyle\sum_i h_{\ell,i}(T) \;+\; m_\ell(T).$$

The next analyses study these vectors directly: the head writes $h_{\ell,i}(T)$ (what each head moves) and the stream states $x_\ell(T)$ (what is carried from layer to layer). They use the same paired design, nulls and independent checks as before.



What we know that the residual analysis can test

- Same-country heads in layers 0–4 (L0H2, L1H11, L3H0, L4H10) attend from the capital to its country at any distance. In layer 0, only the matching capital closes an attention loop.
- The matching capital lowers late-layer resting on BOS at every distance (-0.02 to -0.04 against a wrong capital).
- Previous-token heads look away from an adjacent same-country token. Head behaviour forms families but is ordered by depth only weakly.
- **A structural fact that sets the scope.** Attention is causal, so the country's own stream at τ cannot see T. The match ("this is *its* capital") can only live in the stream at T. The τ stream can carry "a country is here", never "and its capital follows".
- **A built-in null at layer 0.** $x_0(T)$ is just the capital's embedding plus its position. The own and wrong conditions use the same 11 capitals, so their layer-0 clouds are identical by construction. Any difference first appears after layer 0 writes.

Proposed analyses

R1 When does the stream at T know the match?

object $x_\ell(T)$ for $\ell = 0..12$, its capital vs a wrong capital, on paired rows.

method Per layer: persistence of the pooled cloud, and bottleneck distance between the own and wrong clouds against a label-permutation null. A linear probe is the non-topological baseline.

predict Identical at layer 0 (by construction), separable from layer 1 on, after L0H2's write.

counts if The topology shows something a probe does not. For example, own pairs form one tight component while wrong pairs scatter by pair.

R2 Do resting heads write nothing? (the no-op test)

object Each head's write $h_{\ell,i}(T)$ across sequences.

method For each head's write cloud: spread (total persistence, intrinsic dimension, norm), against the head's BOS share, its capital vs a wrong capital.

predict Heads resting on BOS write a near-constant vector (a collapsed cloud). The late heads whose resting drops under the matching capital expand.

counts if BOS share predicts collapse across heads, and the capital's BOS drop comes with expanded writes in the same heads.

R3 What does the stream carry about distance?

object The mean paired shift $m_\ell(\tau) = \mathbb{E}[x_\ell(T \mid \text{country at } \tau) - x_\ell(T \mid \text{control})]$ for $\tau = 1..31$: a 31-point trajectory per layer.

method The trajectory's shape per layer: length, curvature, H0 scale, PCA dimension.

predict Early layers trace a distance-dependent curve; late layers collapse toward one point ("a country is present"). This asks the advisor's local-vs-global question of the content rather than of attention, where we found only a weak depth order.

R4 How does information flow across layers?

object All states (sequence, layer) at T.

method Mapper with layer as the lens and average linkage (single linkage chained in the head analysis). Zigzag persistence across consecutive layers, using the fact that the same sequences appear in every layer.

question Do the own and wrong branches split at some layer and stay split, or merge back (information overwritten)?

R5 Routes, not weights

object A layered graph on (layer, position). Edge weight = attention $\times$ the norm of the value vector actually written (norm-based attention), not raw attention.

method Persistent homology of the flow graph from τ to T: count independent persistent routes, its capital vs a wrong capital.

predict The layer-0 loop survives value weighting only if L0H2 writes a sizeable vector. Otherwise raw attention overstated it.

+ Causal companion: activation patching

Swap $x_\ell(T)$ between the own and wrong runs, layer by layer, and measure late BOS mass, the any-distance heads' attention, and the capital's logit. Topology says where structure appears; patching says where it matters.

Extraction plan (one run, GPT-2 small, CPU)

- TransformerLens hooks:
  - `blocks.`ℓ`.hook_resid_pre` plus `ln_final`, for the stream states;
  - `blocks.`ℓ`.attn.hook_result`, for the per-head writes (needs `model.set_use_attn_result(True)`);
  - `blocks.`ℓ`.hook_mlp_out`, for the MLP writes.

  Store position T only; keep τ and $\tau{+}1$ for the "country present" signal.
- Same cached rows and 11 pairs. Conditions: control, own, wrong, random. $\tau \in \{31, 24, 16, 8, 1\}$ for the point clouds, and all 31 positions for the R3 means.
- Size:
  - stream states: 100 rows $\times$ 11 $\times$ 4 $\times$ 5 = 22,000 sequences $\times$ 13 layers, about 0.45 GB in fp16;
  - head writes: a 30-row subset at 3 positions, about 0.9 GB.

  Minutes of compute; storage is the constraint.

Pitfalls (lessons from the head analyses)

- **Scale.** The residual norm grows with depth, and a few dimensions carry very large values (the attention-sink and massive-activation literature). Normalise per layer, centring and scaling as LayerNorm does, before any distance. Re-check with the top dimensions removed.
- **Dimension.** 768 dimensions against a few thousand points: reduce with PCA to about 50, check stability at 20 / 50 / 100, and bootstrap rows.
- **Degenerate clouds.** The versions of a pair differ only at τ and T, so analyse paired differences, as with attention. Keep the caveat that the same 11 pairs mean consistent, not general.
- **Chaining.** Rips H0 / single linkage made distinctive heads look isolated. Pair persistence with average linkage or kNN communities.
- **Process.** Pre-register each R, and verify each one independently, as before.

To decide together

1. How do we define "what the stream carries": geometry (topology), decodability (probe), or causal effect (patching)? Proposal: all three, with topology leading and the other two as checks.
2. Which positions: T only, where the match can live, plus τ and $\tau{+}1$ for the "country present" signal?
3. What to start with: R2 (the no-op test, closest to the project's core claim) and R1, or R3 (the depth question)?
4. R5's weighting: norm-based (attention $\times$ ||value||) or gradient-based attribution to the capital's logit?

Pointers:

- norm-based attention: Kobayashi et al., 2020;
- attention rollout and flow: Abnar & Zuidema, 2020;
- attention sinks: Xiao et al., 2023; massive activations: Sun et al., 2024;
- zigzag persistence: Carlsson & de Silva, 2010;
- intrinsic dimension across layers: Valeriani et al., 2023; TwoNN: Facco et al., 2017.

Our results: exp4/LOG.md, onepager/exp4b_onepager.pdf, onepager/topology_onepager.pdf.